

Ng Zhengli

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Education

Singapore Polytechnic

Graduated May 2026

Diploma in Information Technology

Relevant Coursework: Data Structures & Algorithms, Front-end Development, Back-end Development, Software Engineering Practices, CI/CD

Skills

Languages: Python, JavaScript, HTML, CSS, SQL

Frameworks & Tools: Flask, React, MongoDB, Playwright, pytest, Git, Figma

Practices: REST APIs, CI/CD, SCRUM, UI/UX Design, Technical Documentation (Sphinx, Docsify)

Projects

Hazardous Substance Routing

Proof of Concept

Python, OpenStreetMap, GeoJSON, VRP

- Reconstructed NEA's approved hazardous substance transport routes as usable GeoJSON by parsing OpenStreetMap data and manually mapping road segments from government-provided spreadsheets
- Implemented a Vehicle Routing Problem solution supporting multiple drivers and delivery destinations with working hour constraints
- Identified and resolved performance bottlenecks that caused runtimes exceeding 5 minutes by rethinking problem structure and algorithm tradeoffs

Chatbot Analytics Dashboard

Internal Tool

React, Flask, MongoDB, Python

- Built a full stack internal dashboard for chatbot monitoring, covering authentication, user management, lead tracking, and profile pages
- Worked with MongoDB for the first time, having previously only used SQL databases
- Identified gaps in the design-to-development handoff process, advocating for clearer design specs to reduce repeated CSS refactors
- Implemented reusable React components to improve maintainability across the frontend

Experience

UBTS Pte Ltd

Junior Tech Developer

Apr 2025 – Jun 2026

- Researched and prototyped a routing solution for hazardous substance transportation, mapping government-sourced road restrictions onto OpenStreetMap data, cleaning and manually editing GeoJSON to ensure routing accuracy
- Developed and maintained a full stack analytics dashboard for chatbot monitoring, using Flask, React, and MongoDB
- Migrated RPA workflows from ElectroNeek to an open source solution, eliminating enterprise software licensing costs
- Migrated a chatbot backend from Java to Python, analysing and rewriting existing codebase to improve maintainability

Achievements & Activities

Awards

- Edusave Certificate of Academic Achievement – Ministry of Education Singapore, 2025
- Sembawang West CCC – North West CDC Education Merit Award

Volunteer Work

- Project Rice, Save the Wave, Meals on Wheels